ENVA — 100 Million Trees Initiative

Evidence-Based Smart Afforestation Planning System

Study Area: Kafr El Dawwar City — CITY ONLY

Run ID: ENVA_100MT_20260906_012208

1. Executive Summary

This report presents the outputs of the independent ENVA 100 Million Trees Initiative planning workflow for Kafr El Dawwar City. The workflow uses the verified city-only boundary, OpenStreetMap road geometry, satellite-derived vegetation context, geometric planting feasibility, a transparent priority model, species recommendation logic, and indicative water/cost calculations.

The outputs are a planning prototype and should not be interpreted as a final field-approved planting schedule. Physical site inspection, utility checks, irrigation review, species validation, and competent local authority approval remain necessary before implementation.

Metric	Value	Status
Proposed trees	23,000	■ Derived
Road segments	6,454	■ Measured
Eligible planting length	179.96 km	■ Derived
Estimated total cost	11,500,000 EGP	■ Estimated
Estimated daily water demand	345,000 L/day	■ Estimated
High-priority streets	1,710	■ Derived

2. Study Area and Governance

The analysis is explicitly restricted to Kafr El Dawwar City. The Markaz is excluded from this workflow. The same verified city-only boundary was reused as a read-only source reference in Cell 2.

3. Data and Methodology

Component	Source / Method	Status
Boundary	Verified ENVA city-only AOI	Measured / Verified
Road network	OpenStreetMap via osmnx + GIS clipping	Measured
Vegetation context	Sentinel-2 SR Harmonized NDVI	Derived
Planting feasibility	Road geometry + exclusion screening	Derived
Priority	Heat context + vegetation deficit + feasibility	Prototype / Derived
Species recommendation	Multi-factor candidate scoring	Prototype
Water	Indicative species demand categories	Estimated
Cost	Tree count × prototype unit cost	Estimated

4. Priority Model

The street-priority prototype combines three available factors: heat context (35%), vegetation deficit (35%), and planting feasibility (30%). The weighting is explicitly a prototype assumption requiring methodological review. Population exposure was not included because a verified population-density dataset was not available in this run.

Class	Street count
High	1,710
Medium	4,726
Low	18

5. Water Demand

The current model estimates approximately 345,000 liters/day and 10,350,000 liters per 30-day month for the generated tree portfolio. These figures are indicative planning estimates and are not operational irrigation allocations. Actual demand depends on species, age, season, soil, irrigation method, survival strategy, and local management practice.

6. Cost

Using the current prototype assumption of 500 EGP per proposed tree, the modeled tree portfolio corresponds to an estimated 11,500,000 EGP. This is not a procurement quote and should be replaced by approved local procurement prices and associated implementation costs before financial approval.

7. Evidence and Confidence Framework

Class	Meaning
Measured	Directly measured or directly calculated from available source geometry/data.
Derived	Computed deterministically from measured/source data using a stated method.
Estimated	Estimated using an explicit assumption or indicative parameter.
Prototype	Planning/model parameter requiring methodological or local validation.
Field Validation Required	Result should not be treated as a final field-approved planting decision until physical site verification and local authority review

8. Key Limitations and Required Validation

• Tree locations are model-generated planning points, not surveyed planting stakes.

• Road and place attributes depend on OpenStreetMap coverage and tagging.

• NDVI is vegetation-index context and does not equal a field tree inventory.

• Species suitability is a prototype recommendation and requires local validation.

• Water demand uses indicative assumptions and should be calibrated to actual irrigation practice.

• Tree spacing, intersection buffer, and minimum planting segment thresholds are prototype parameters.

• The priority weighting requires methodological review before institutional adoption.

• Underground utilities and physical site constraints require field verification.

9. Decision-Maker Conclusion

The system provides an auditable planning framework for identifying candidate tree locations and prioritizing streets within the approved city boundary. The strongest value of the present release is not a claim of final planting completion, but a reproducible evidence chain from verified boundary → road geometry → feasibility → candidate trees → priority → indicative water/cost estimates. Final implementation should proceed only after the stated validation steps are completed.